



DATE: May 9, 2026

TO: [REDACTED]

FROM: Jett Palmer, Statistics

SUBJECT: Investigating the Associations Between Soil Health, Land Management Practices, and Grower-Perceived Outcomes

Executive Summary

This report investigates the relationships between soil health indicators (SHIs), land management practices (LMPs), and grower-perceived outcomes (GPOs) in California vineyards to assess whether sustainable practices align with productive agricultural outcomes.

Using data from 83 vineyard blocks across 11 vineyards, the analysis combined **multiple correspondence analysis (MCA)** and **ordinal generalized linear mixed models (GLMMs)** to identify patterns and test associations. MCA was used to uncover latent structures in grower-reported outcomes and to identify the most informative variables, while GLMMs evaluated relationships between soil health, management practices, and key outcomes, accounting for vineyard-level variation.

Key Findings

- The first two MCA dimensions explained **19.3% of variability** in grower-perceived outcomes, highlighting moderate but meaningful structure in the data.
- Variables related to **synthetic input use (nitrogen, potassium, phosphorus)**, **pest pressures**, and **soil organic matter** were the most influential in explaining variation in outcomes (Table 7).
- Regression results suggested directional relationships between practices and outcomes:
 - **Grazing** was associated with higher odds of increased synthetic nitrogen and potassium use, but lower disease pressure.
 - **Composting** was associated with slightly higher insect plant pressure.
 - Higher **vine soil health scores** were strongly associated with lower insect plant pressure.